



UITs

**UNIVERSITY OF INFORMATION
TECHNOLOGY AND SCIENCES**

Lab Report

Course Title: Internet of Things Lab

Course Code: CSE 402

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Experiment no: 03

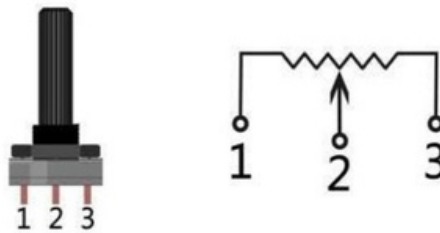
Experiment Name: Controlling the LED blink rate with the potentiometer interfacing with Arduino

Objective:

To use the potentiometer to control the blink rate of a single LED.

Introduction:

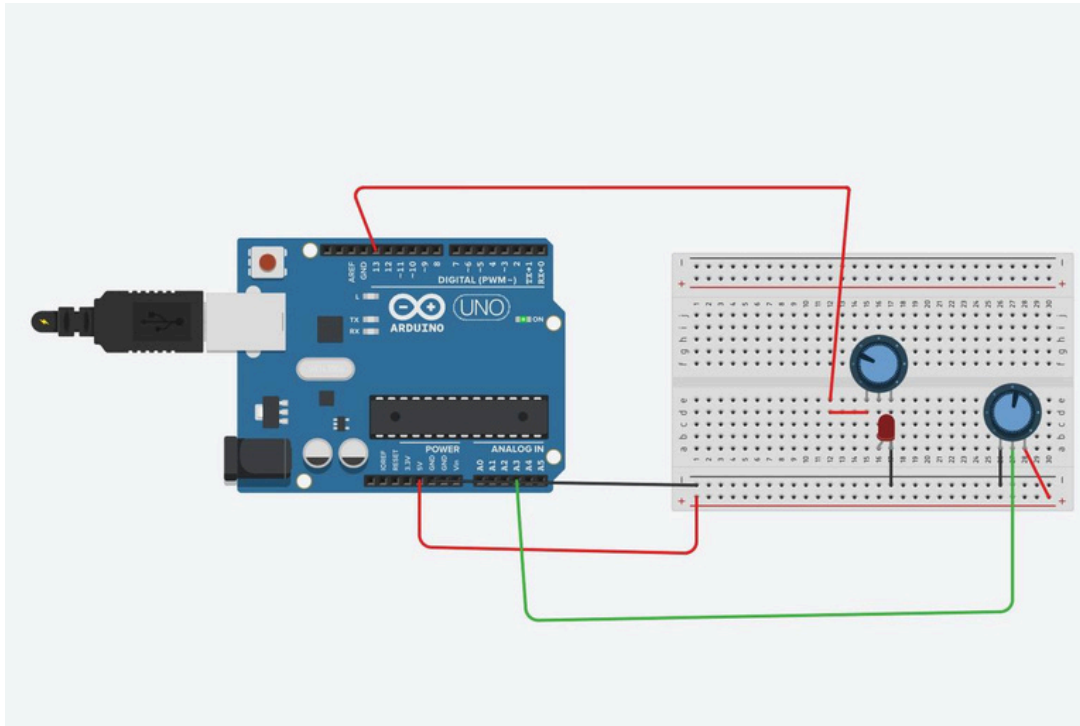
A potentiometer is a variable resistor with a knob that allows altering the resistance of the potentiometer. The potentiometer manipulates a continuous analog signal, which represents physical measurements. The potentiometer is used with Arduino to control the blink rate of the LED. The potentiometer is an adjustable resistor, and its operating principle is shown in the following figure:



Hardware Required:

Component	Quantity
Arduino Uno	1
Bread board	1
220 Ω current limiting resistor	1
5mm LED	1
10K Ω Potentiometer	1
Jumper Wires	Several
Supporting USB data cable	1

Circuit Diagram:



Working Procedure:

1. Insert the potentiometer into your breadboard and connect its center pin to the analog pin A2 and the remaining pin to GND on the breadboard.
2. Insert the LED into the breadboard. Attach the positive leg (the longer leg) to pin 13 of the Arduino via the 220-ohm resistor, and the negative leg to GND.
3. Upload the code as given below.
4. Turn the potentiometer to control the brightness of the LED and move the position of pin 2 by rotating the knob, changing the resistance value from pin 2 to both ends
5. Observe the changes in the blinking rate of the LED.

Code:

```
const int regPin=A3;

const int LedPin =13;

int Value=0;

void setup()

{

    pinMode(regPin, INPUT);

    pinMode(LedPin, OUTPUT);

}

void loop()

{

    Value=analogRead (regPin);

    digitalWrite(LedPin,HIGH);

    delay(Value);

    digitalWrite(LedPin,LOW);

    delay(Value);

}
```

Conclusion:

This experiment successfully demonstrated the control of LED blink rate using a potentiometer interfaced with an Arduino microcontroller. By varying the potentiometer's position, the analog input to the Arduino was altered, directly influencing the timing of the LED's on/off cycles. This confirms the principle of analog-to-digital conversion and its application in controlling real-world devices through programmed logic. The experiment highlights the versatility of microcontrollers in creating interactive systems and provides a foundational understanding of sensor input and actuator control.